Arithmetic Board 1

File: AU1.kicad_sch

Arithmetic Board 2

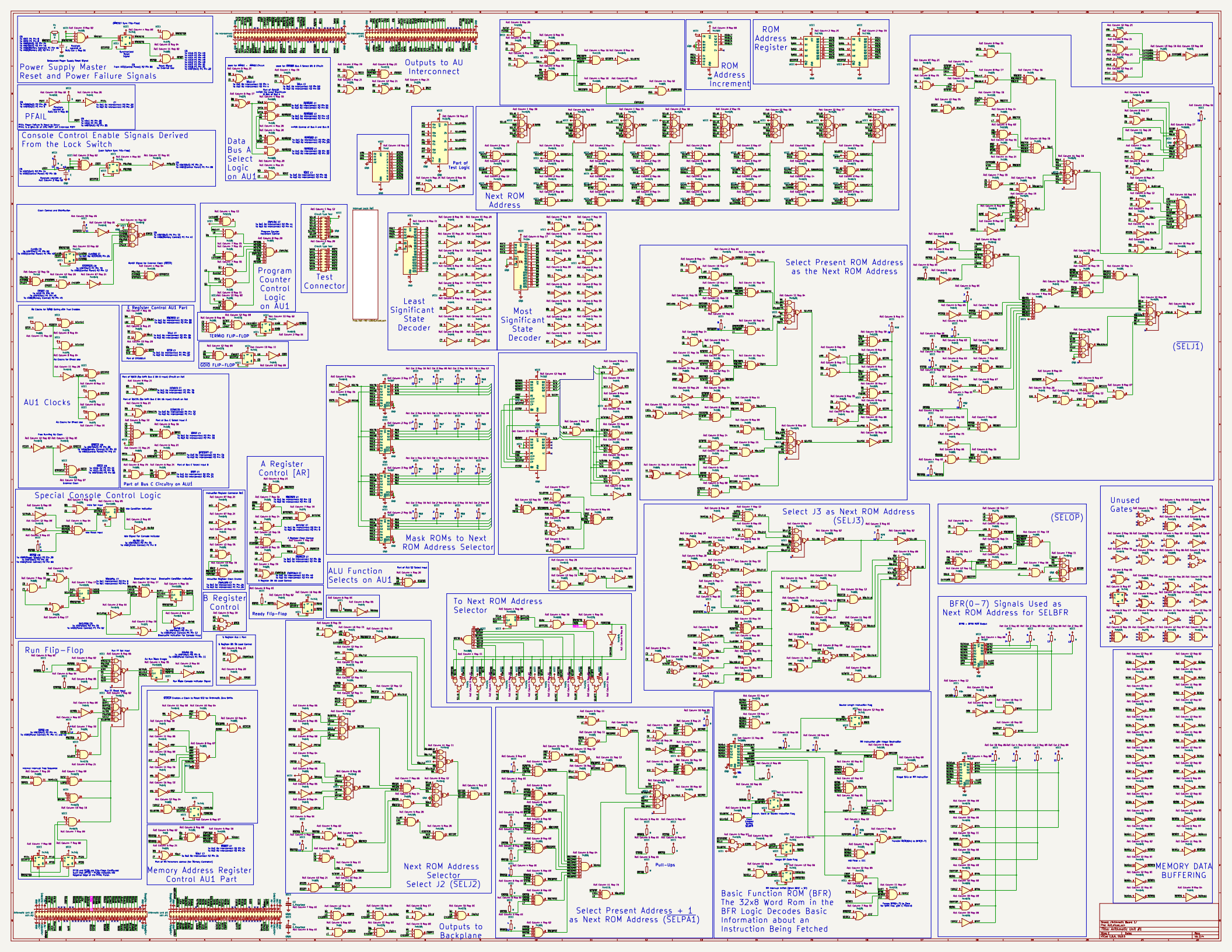
File: AU2.kicad_sch

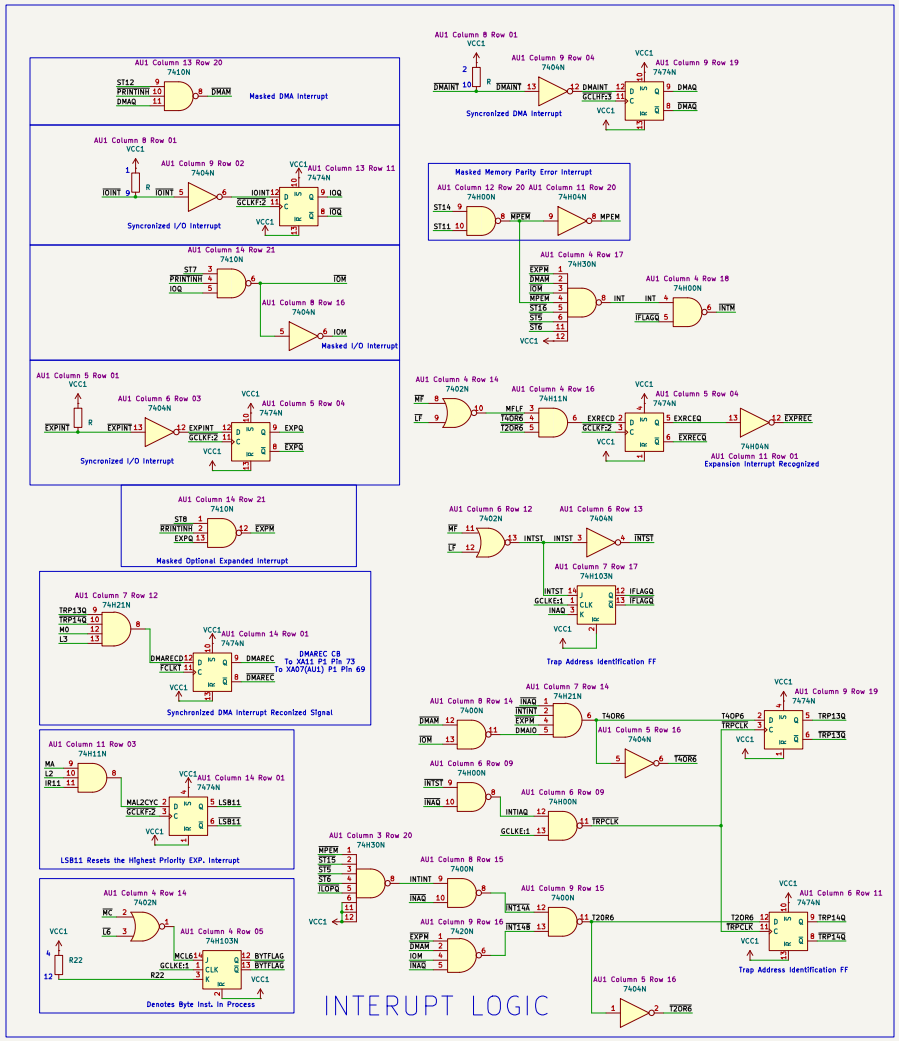
Derived from Maintenance Manual VIII & IX
TI980A Arithmetic Unit
Texas Instruments

Sheet: /
File: TI980A-Arithmetic-Unit.kicad_sch

Title: 960713-9901

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 1/4

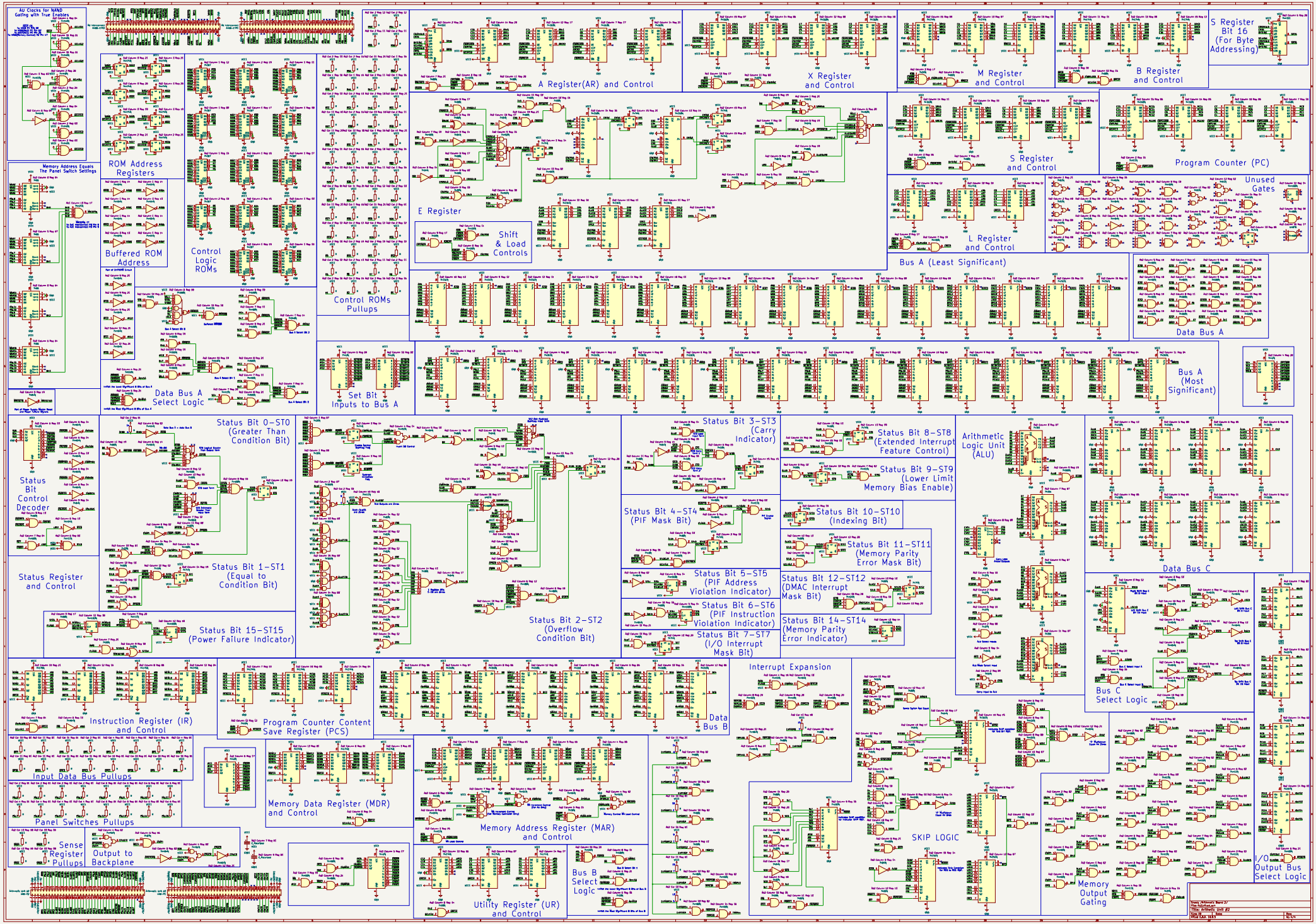

Power Supply Master Reset and Power Failure Signals
PFAIL
Console Control Enable Signals Derived From the Lock Switch
Data Bus A Select Logic on AU1
Outputs to AU Interconnect
Part of Test Logic
Next ROM Address
ROM Address Increment
ROM Address Register
Program Counter Control Logic on AU1
Test Connector
Least Significant State Decoder
Most Significant State Decoder
Select Present ROM Address as the Next ROM Address
(SELJ1)
E Register Control AU1 Part
TERMIO FLIP-FLOP
GOTO FLIP-FLOP
AU1 Clocks
Mask ROMs to Next ROM Address Selector
A Register Control [AR]
Special Console Control Logic
ALU Function Selects on AU1
Ready Flip-Flop
B Register Control
To Next ROM Address Selector
Select J3 as Next ROM Address (SELJ3)
(SELOP)
Unused Gates
BFR(0-7) Signals Used as Next ROM Address for SELBFR
Run Flip-Flop
Memory Address Register Control AU1 Part
Next ROM Address Selector Select J2 (SELJ2)
Pull-Ups
Select Present Address + 1 as Next ROM Address (SELPA1)
Basic Function ROM (BFR) The 32x8 Word Rom in the BFR Logic Decodes Basic Information about an Instruction Being Fetched
MEMORY DATA BUFFERING
Outputs to Backplane



Sheet: /Arithmetic Board 1/Interrupt Logic AU1/
File: AU1-INT-LOGIC.kicad_sch

Title: AU1 Interrupt Logic

Size: C	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 3/4


AU Clocks for NAND Gating with True Enables
Memory Address Equals The Panel Switch Settings
ROM Address Registers
Buffered ROM Address
Control Logic ROMs
Control ROMs Pullups
Data Bus A Select Logic
Set Bit Inputs to Bus A
A Register(AR) and Control
E Register
Shift & Load Controls
X Register and Control
M Register and Control
B Register and Control
S Register Bit 16 (For Byte Addressing)
S Register and Control
Program Counter (PC)
L Register and Control
Unused Gates
Bus A (Least Significant)
Data Bus A
Bus A (Most Significant)
Status Register and Control
Status Bit Control Decoder
Status Bit 0-ST0 (Greater Than Condition Bit)
Status Bit 1-ST1 (Equal to Condition Bit)
Status Bit 2-ST2 (Overflow Condition Bit)
Status Bit 3-ST3 (Carry Indicator)
Status Bit 4-ST4 (PIF Mask Bit)
Status Bit 5-ST5 (PIF Address Violation Indicator)
Status Bit 6-ST6 (PIF Instruction Violation Indicator)
Status Bit 7-ST7 (I/O Interrupt Mask Bit)
Status Bit 8-ST8 (Extended Interrupt Feature Control)
Status Bit 9-ST9 (Lower Limit Memory Bias Enable)
Status Bit 10-ST10 (Indexing Bit)
Status Bit 11-ST11 (Memory Parity Error Mask Bit)
Status Bit 12-ST12 (DMAC Interrupt Mask Bit)
Status Bit 14-ST14 (Memory Parity Error Indicator)
Status Bit 15-ST15 (Power Failure Indicator)
Arithmetic Logic Unit (ALU)
Data Bus C
Bus C Select Logic
Instruction Register (IR) and Control
Program Counter Content Save Register (PCS)
Data Bus B
Interrupt Expansion
Input Data Bus Pullups
Memory Data Register (MDR) and Control
Memory Address Register (MAR) and Control
Panel Switches Pullups
Sense Register Pullups
Output to Backplane
Bus B Select Logic
Utility Register (UR) and Control
SKIP LOGIC
Memory Output Gating
I/O Output Bus Select Logic